

DEXWEAVE: LEARNING DEXTEROUS HUMANOID LOCO-MANIPULATION FROM HUMAN DEMONSTRATIONS

Naichuan Sun^{1,2*} Haotian Shen^{1*} Yizhang Zhang¹ Luying Feng¹ Haoze Wang¹
Yuanbo Xiangli² Yaochu Jin¹ Peidong Liu^{1†}

¹Department of Artificial Intelligence, School of Engineering, Westlake University

²School of Artificial Intelligence, Shanghai Jiao Tong University

naichuan.sun@sjtu.edu.cn

{shenhaotian, liupeidong}@westlake.edu.cn



Figure 1: **DexWeave** translates human demonstrations into executable dexterous humanoid loco-manipulation skills.

ABSTRACT

Learning dexterous humanoid loco-manipulation from human demonstrations requires transferring not only human motion, but also the coordinated interaction structure underlying the demonstrated behavior. This is challenging because embodiment differences distort the coupling among body motion, wrist placement, finger articulation, and object interaction, while kinematically accurate references may still be difficult to realize under robot dynamics. We present **DexWeave**, a unified framework that connects interaction-consistent motion retargeting with anatomy-aware whole-body policy learning. DexWeave first employs a two-stage retargeting procedure that initializes body and hand motions with specialized solvers and subsequently performs coupled refinement over the upper-body interaction chain while preserving lower-body support. The resulting references are tracked by an anatomy-aware Transformer policy that represents anatomical regions as structured tokens and uses directed masked attention to model their dependencies, with object information selectively conditioning the upper-body pathway for dexterous interaction. The policy jointly outputs body and dexterous-hand actions and is trained directly with reinforcement learning, without pretrained tracking policies, teacher-student distillation, or subsequent residual refinement. DexWeave improves retargeting fidelity and interaction consistency while achieving higher manipulation performance and faster policy convergence than MLP baselines. We further deploy the learned policies on a physical Unitree G1 humanoid equipped with Inspire dexterous hands, demonstrating dexterous whole-body loco-manipulation in the real world. See our project page for videos.

*Equal contribution.

†Corresponding author.

1 INTRODUCTION

Dexterous humanoid loco-manipulation could enable a wide range of real-world tasks, but requires tight coordination between locomotion, whole-body support, and articulated hand-object interaction. Learning from demonstrations has emerged as a scalable approach to acquiring robot manipulation and humanoid whole-body skills (Chi et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Allshire et al., 2025; Heng et al., 2026). In particular, abundant human motion and video data provide rich demonstrations of coordinated whole-body behaviors without requiring costly robot teleoperation (He et al., 2025; Heng et al., 2026; Allshire et al., 2025; Jiang et al., 2026). Leveraging such data for dexterous humanoids, however, requires retargeting both body motion and hand articulation to the robot’s morphology while preserving the demonstrated interactions, together with a control policy that can realize these interactions under robot dynamics.

A common challenge underlies both retargeting and control: wrist placement and finger articulation jointly determine hand-object interaction, yet they are often optimized at different levels of the motion hierarchy. Whole-body retargeting approaches improve motion fidelity and kinematic feasibility, while dexterous-hand methods primarily optimize hand-level kinematics and local hand-object geometry (Handa et al., 2020; Xin et al., 2026; Wu et al., 2026). Despite advances in teleoperation and physics-based refinement (Heng et al., 2026; Pan et al., 2025), these levels can remain weakly coupled: differences in arm reach, palm geometry, and finger dimensions may make wrist placement and finger articulation mutually inconsistent even when each is locally well retargeted. The same coupling reappears during control: manipulation forces perturb whole-body balance, while locomotion continuously changes wrist placement and can disrupt hand-object contact. Existing policies address whole-body coordination through jointly trained upper- and lower-body agents (Zhang et al., 2026a), unified policies for locomotion and upper-body control (Sun et al., 2025), or pretrained body and hand priors coordinated through a residual policy (Li et al., 2026a). While these approaches make high-dimensional control more tractable, fine-grained coordination between whole-body motion and articulated hand interaction remains challenging. Our key insight is that successful human-to-humanoid dexterous skill transfer requires preserving demonstrated interaction structure across embodiments and exploiting anatomical structure as an inductive bias for whole-body policy learning.

Guided by this insight, we present **DexWeave**, an integrated framework that bridges interaction-consistent motion retargeting and anatomy-aware whole-body policy learning. A two-stage retargeting pipeline first initializes robot motion using specialized body and hand solvers with contact-aware wrist blending, and then jointly refines the arms, wrists, and fingers along the upper-body interaction chain to preserve wrist-finger-object coordination while maintaining feasible lower-body support. An anatomy-aware Transformer organizes whole-body control through regional body tokens and directed masked attention that captures structured dependencies across anatomical regions. Object information is selectively injected into the upper-body pathway, conditioning dexterous interaction while leaving the lower-body pathway unconditioned on direct object observations. For each reference motion, a single policy network jointly outputs body and dexterous-hand actions and is trained with Proximal Policy Optimization (PPO), without tracking-policy pretraining, teacher-student distillation, or subsequent residual refinement.

We evaluate DexWeave primarily on whole-body interactions involving fine-grained dexterous hand-object contact from GRAB (Taheri et al., 2020) and HUMOTO (Lu et al., 2025), with additional evaluations on whole-body object interaction from OMOMO (Li et al., 2023) and body-only motion from LAFAN1 (Harvey et al., 2020). Retargeting comparisons show improved kinematic feasibility, with reduced penetration and foot skating, while more accurately preserving hand-object interactions. For closed-loop control, evaluations on representative dexterous loco-manipulation motions show that our policy achieves higher manipulation performance and approximately $2\times$ faster convergence than MLP baselines. We further deploy DexWeave on a physical Unitree G1 humanoid equipped with Inspire dexterous hands, demonstrating dexterous whole-body loco-manipulation in the real world. Beyond the policies studied here, DexWeave provides a potential pathway for converting abundant human interaction data into embodiment-specific, physically executable robot data for future embodied foundation models.

2 RELATED WORK

Learning Humanoid Skills from Human Demonstrations. Human demonstrations provide a scalable source of whole-body behaviors without requiring extensive robot-specific teleoperation. OmniH2O (He et al., 2025) combines large-scale human motion retargeting with reinforcement learning to acquire whole-body humanoid skills, while VideoMimic (Allshire et al., 2025) reconstructs humans and their surrounding environments from videos to learn context-aware humanoid control. More recent systems extend this direction toward whole-body manipulation. HumDex (Heng et al., 2026) leverages whole-body human motion to learn transferable motion priors before adapting them with robot data. These works demonstrate the potential of human data for scalable humanoid skill acquisition. DexWeave focuses on a complementary challenge: transferring demonstrations with coordinated body, hand, and object motion across embodiments while preserving the interactions needed for physically executable dexterous whole-body skills.

Human-to-Robot Motion Retargeting. Motion retargeting bridges the substantial morphology gap between human demonstrations and humanoid robots. Whole-body methods adapt human motion to robot kinematics while explicitly enforcing constraints such as joint limits, collisions, contacts, and support feasibility (Araujo et al., 2026; Yang et al., 2026; NVIDIA, 2026). In parallel, dexterous-hand methods focus on transferring fine-grained finger articulation and hand–object relationships (Handa et al., 2020; Xin et al., 2026; Wu et al., 2026). TopoRetarget (Wu et al., 2026), for example, explicitly preserves task-relevant hand–object interaction structure, while SPIDER (Pan et al., 2025) employs physics-informed refinement to transform kinematic human demonstrations into dynamically feasible robot trajectories. Despite this progress, whole-body and dexterous-hand retargeting are often optimized with different objectives or at separate stages. Across embodiments, differences in arm reach, palm geometry, and finger proportions can therefore make wrist placement and finger articulation mutually inconsistent even when each is locally well retargeted. This motivates preserving interaction structure across the entire upper-body interaction chain rather than only at the body or hand level. DexWeave jointly refines the arms, wrists, and fingers to maintain wrist–finger–object coordination while preserving feasible lower-body support.

Humanoid Loco-Manipulation. Humanoid loco-manipulation requires coordinating locomotion, balance, upper-body motion, and object interaction within a high-dimensional action space. Existing approaches make this problem tractable through control decomposition, unified whole-body policies, or learned motion priors. FALCON (Zhang et al., 2026a) decomposes force-adaptive loco-manipulation into jointly trained lower-body locomotion and upper-body manipulation agents. ULC (Sun et al., 2025) demonstrates unified locomotion and upper-body control within a single policy, while CoordDex (Li et al., 2026a) extends loco-manipulation to dexterous hands using separately trained and distilled body and hand priors coordinated through downstream residual reinforcement learning. More recently, vision–language–action (VLA) and world-action models (WAMs) have extended humanoid loco-manipulation toward generalizable visuomotor control. WholeBodyVLA (Jiang et al., 2026) leverages action-free human videos to alleviate the scarcity of humanoid loco-manipulation data, while OpenHLM (Hu et al., 2026) maps visual and language observations to whole-body humanoid actions. MotionWAM (Zheng et al., 2026), ω -0 (Li et al., 2026b), and related work (Li et al., 2026c) further exploit large-scale visual or motion priors for whole-body action generation. These developments expose a complementary data bottleneck: human motion and interaction data are abundant, yet embodiment-specific, physically executable whole-body dexterous trajectories remain costly to acquire. DexWeave addresses this conversion gap by grounding coordinated human body–hand–object demonstrations into interaction-consistent, physically executable robot behaviors, providing a complementary source of whole-body supervision for future embodied models.

3 DEXWEAVE

3.1 OVERVIEW

Given a human body–hand–object demonstration, DexWeave converts it into a physically executable dexterous humanoid skill by preserving interaction structure across embodiments and exploiting anatomical structure for control. As shown in Fig. 2, the framework first constructs an interaction-

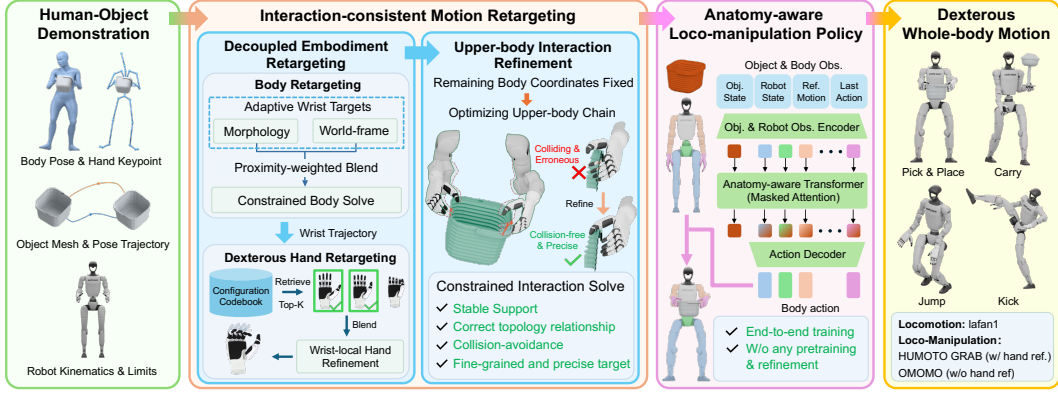


Figure 2: **Overview of DexWeave.** Given a human body–hand–object demonstration, DexWeave first constructs an interaction-consistent robot reference through specialized body–hand initialization and coupled refinement of the upper-body interaction chain. An anatomy-aware whole-body loco-manipulation policy then realizes the reference under robot dynamics using regional anatomical tokens, directed masked attention, and selective object conditioning, producing executable dexterous humanoid loco-manipulation skills.

consistent robot reference and then learns a policy to execute it under physical robot dynamics. **Interaction-consistent motion retargeting** uses specialized body–hand initialization and coupled upper-body refinement to preserve hand–object relationships while maintaining the established support motion (Sec. 3.2). **Anatomy-aware loco-manipulation policy learning** then uses regional anatomical tokens, directed information flow, and selective object conditioning to coordinate locomotion and dexterous manipulation (Sec. 3.3).

3.2 INTERACTION-CONSISTENT MOTION RETARGETING

Retargeting human demonstrations to a robot must reconcile whole-body reachability, hand–object contact fidelity, and physical feasibility. Optimizing all three jointly is ill-conditioned, because body-scale differences, finger-dimension mismatches, and collision constraints interact across the full kinematic chain. DexWeave therefore decomposes the problem according to interaction locality. A decoupled initialization stage first establishes body motion and hand shape where each is locally well-posed (Sec. 3.2.1), and a subsequent refinement stage closes the residual misalignment on the compact upper-limb chain that mediates contact (Sec. 3.2.2).

A demonstration provides human body poses $\mathbf{X}_t^H$, hand keypoints $\mathbf{H}_t^s$ for $s \in \{L, R\}$, and rigid-object poses $\mathbf{X}_t^O$ over frames $t = 1, \dots, T$. The robot configuration $\mathbf{q}_t = (\mathbf{q}_t^B, \boldsymbol{\theta}_t^L, \boldsymbol{\theta}_t^R)$ consists of the floating base and non-finger joints $\mathbf{q}_t^B$ together with the independent hand drivers $\boldsymbol{\theta}_t^s$, from which dependent finger joints are determined by the robot’s mimic relations. The pipeline first constructs a decoupled initialization $\bar{\mathbf{q}}_{1:T}$ and then refines its upper-body coordinates to produce $\mathbf{q}_{1:T}^*$. Together with the original object trajectory, this yields the reference $\mathcal{R}_{1:T} = \{(\mathbf{q}_t^*, \mathbf{X}_t^O)\}_{t=1}^T$ for downstream policy learning.

3.2.1 DECOUPLED BODY–HAND INITIALIZATION

The initialization stage solves body motion and hand articulation using specialized solvers before coupling them during interaction refinement. This decomposition reflects the different structure of the two subproblems: body retargeting is dominated by large-scale morphology adaptation and support constraints, whereas hand fitting depends on local finger geometry and kinematic coupling. Solving each component in the representation where it is locally well posed provides a robust initialization for subsequent joint refinement.

Contact-Aware Body Retargeting. The wrist is the point at which whole-body retargeting most directly affects manipulation. Rescaling human motion to robot proportions can displace the wrist relative to nearby objects and break contact even when the overall body pose is well retargeted. We therefore blend two wrist targets: a robot-scaled morphology-preserving target (Araujo et al., 2026)

and a world-aligned interaction-preserving target. Their contribution is controlled by a per-hand weight $\alpha_{t,s} \in [0, 1]$ determined by persistent hand–object proximity. When the hand is far from the object, the target favors morphology adaptation; as sustained interaction emerges, the blend gradually shifts toward scene alignment. This provides a smooth transition without introducing a discrete contact mode switch. Wrist orientations remain fully mapped and the object trajectory is kept unchanged. Support constraints inferred from source foot kinematics suppress unintended foot sliding, while clearance penalties handle self-, ground-, and body–object collisions. At each frame, the body configuration is obtained by solving:

$$\bar{\mathbf{q}}_t^B = \arg \min_{\mathbf{q}^B \in \mathcal{Q}_t^B} \lambda_{\text{track}} \mathcal{L}_{\text{track}} + \lambda_{\text{clr}} \mathcal{L}_{\text{clr}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}} + \mathcal{L}_{\text{reg}}, \quad (1)$$

where $\mathcal{L}_{\text{track}}$ penalizes deviation from body landmarks and the blended wrist targets $\widehat{\mathcal{W}}_t$, $\mathcal{L}_{\text{clr}}$ collects soft collision terms, $\mathcal{L}_{\text{temp}}$ regularizes temporal posture, and $\mathcal{L}_{\text{reg}}$ denotes a regularization term. The feasible set $\mathcal{Q}_t^B$ encodes joint limits, support equalities, and body–collision boundaries.

Retrieval-Based Dexterous Hand Fitting. Nonlinear finger kinematics and mimic coupling make direct geometric fitting sensitive to initialization (Handa et al., 2020; Xin et al., 2026). We address this using a precomputed codebook $\mathcal{B}^s = (\boldsymbol{\theta}_n^s, \mathbf{g}_n^{s,R})_{n=1}^{N_B}$, which pairs feasible hand-driver configurations with wrist-local geometric descriptors. These descriptors encode fingertip positions, chain directions, and inter-finger relations computed through forward kinematics. Given observed human hand keypoints, we construct the corresponding descriptor $\mathbf{g}_t^{s,H}$ in the same wrist-local frame and retrieve the K nearest codebook entries. Their driver configurations are blended using normalized distance weights to obtain $\widehat{\boldsymbol{\theta}}_t^s$, which serves as both a warm start and a soft anchor for continuous fitting of the independent drivers within their mimic-consistent bounds. Sequences without articulated hand references default to a neutral configuration $\boldsymbol{\theta}_{\text{neutral}}^s$. Combining the fitted hands with the retargeted body trajectory yields the whole-body initialization $\bar{\mathbf{q}}_{1:T}$.

3.2.2 COUPLED UPPER-BODY INTERACTION REFINEMENT

The previous initialization can still exhibit residual hand–object misalignment because embodiment differences propagate jointly through arm reach, palm geometry, and finger proportions. Correcting such errors only at the wrist or fingers can therefore distort another part of the interaction chain. DexWeave instead jointly refines the upper-body kinematic chain that governs hand–object interaction. In particular, we define a compact set of joint variables $\mathbf{z}_t = [\mathbf{q}_t^U, \boldsymbol{\theta}_t^L, \boldsymbol{\theta}_t^R]^\top$ that contains the bilateral arm and wrist joints (i.e. $\mathbf{q}_t^U$) together with both hands’ independent drivers (i.e. $\boldsymbol{\theta}_t^L, \boldsymbol{\theta}_t^R$). All other body joints retain their initialized values in $\bar{\mathbf{q}}_t$, preserving the lower-body support motion.

Three design choices shape this refinement. (i) *Compact active set.* Restricting optimization to the upper-body interaction chain confines corrections to joints that directly influence contact and prevents unnecessary drift in lower-body support. (ii) *Priority-weighted fingertip alignment.* Following precision-grasping principles (Handa et al., 2020), we assign larger weights to the thumb and index fingertips, which often dominate interaction geometry, while the remaining fingertips provide complementary grasp and support cues. (iii) *Relaxed wrist anchoring.* Rather than fixing the wrist at its initialized position, we soften its positional anchor while retaining palm-orientation objectives (Xin et al., 2026). This allows residual mismatch to be distributed across wrist placement and finger articulation instead of forcing the fingers alone to absorb the correction.

Let $\bar{\mathbf{z}}_t$ denote the initialized active coordinates. At each frame, the refinement solves

$$\mathbf{z}_t^* = \arg \min_{\mathbf{z} \in \mathcal{Z}_t} \lambda_{\text{tip}} \mathcal{L}_{\text{tip}} + \lambda_{\text{ori}} \mathcal{L}_{\text{ori}} + \lambda_{\text{clr}} \mathcal{L}_{\text{clr}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}} + \mathcal{L}_{\text{reg}}, \quad (2)$$

where $\mathcal{L}_{\text{tip}}$ penalizes priority-weighted fingertip position error, $\mathcal{L}_{\text{ori}}$ tracks the palm normal, wrist-forward direction, and hand-lateral direction, $\mathcal{L}_{\text{clr}}$ collects soft collision-clearance and recovery penalties, $\mathcal{L}_{\text{temp}}$ regularizes the correction $\mathbf{z}_t - \bar{\mathbf{z}}_t$ over time, and $\mathcal{L}_{\text{reg}}$ denotes a regularization term. Replacing the active joint values of $\bar{\mathbf{q}}_t$ with $\mathbf{z}_t^*$ yields the refined whole-body trajectory $\mathbf{q}_{1:T}^*$. Through this coupled refinement, wrist placement and finger articulation are optimized as a single interaction structure rather than as independent targets. Loss decompositions, dataset-specific target assignments, and the complete definitions of $\mathcal{Q}_t^B$ and $\mathcal{Z}_t$ are provided in Appendix A.

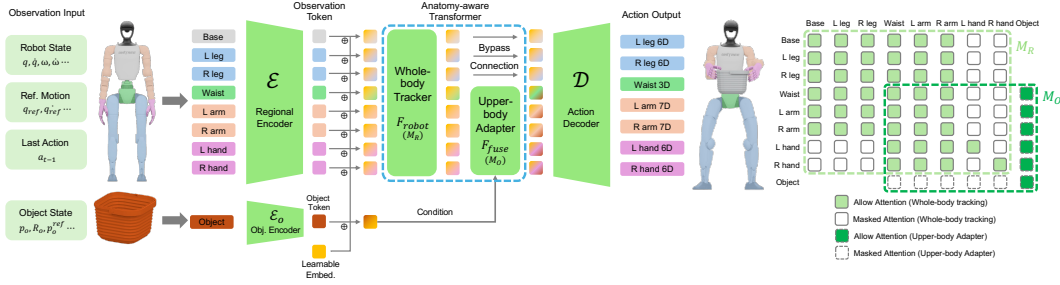


Figure 3: **Anatomy-aware policy architecture.** **Left:** Robot observations are factorized into anatomical tokens and coordinated by a masked whole-body Transformer. Object information is fused one-way into the waist, arm, and hand branches, while the original waist feature is retained for object-independent leg control. Seven group-specific action heads produce the full body-and-hand action. **Right:** Attention-mask visualization. Rows attend to columns. Light and dark green indicate allowed attention in the robot Transformer and object-fusion Transformer, respectively.

3.3 ANATOMY-AWARE LOCO-MANIPULATION POLICY LEARNING

Retargeting provides interaction-consistent references, but realizing them under robot dynamics requires coordinated control of locomotion, whole-body posture, and articulated hands. DexWeave explicitly incorporates humanoid anatomy into both representation and information flow using an asymmetric actor-critic architecture. The actor receives a 238-dimensional observation $\mathbf{o}_t = [\mathbf{o}_t^{\text{robot}}, \mathbf{o}_t^{\text{object}}]$, comprising a 211-dimensional robot observation and a 27-dimensional object observation, while the critic uses privileged simulation observations during training. The network architecture is shown in Figure 3.

3.3.1 REGIONAL TOKENIZATION AND DIRECTED COORDINATION

The robot observation is factorized into eight anatomical regions corresponding to the base, bilateral legs, waist, bilateral arms, and bilateral hands. Region-specific encoders $\mathcal{E}_i$ map each regional observation $\mathbf{x}_i$ into a shared d -dimensional space ($d = 128$), with a learned body-part embedding $\mathbf{p}_i$:

$$\mathbf{h}_i = \mathcal{E}_i(\mathbf{x}_i) + \mathbf{p}_i, \quad \mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_8]^\top. \quad (3)$$

A two-layer masked Transformer produces

$$\mathbf{Z} = F_{\text{robot}}(\mathbf{H}; M_R), \quad (4)$$

where the anatomical mask M_R allows bidirectional communication among the base, legs, waist, and arms, while preventing body tokens from attending to the hands. Each hand token attends to the waist, both arms, and itself. This directed structure provides finger control with upstream body context without feeding hand-specific features back into the body-control representation.

3.3.2 SELECTIVE OBJECT CONDITIONING

The object observation contains the current pose, reference pose, and their discrepancy in the torso frame, where each pose uses a 3D position and a 6D rotation representation from the first two columns of the rotation matrix. It is encoded as

$$\mathbf{h}_o = \mathcal{E}_o(\mathbf{o}_t^{\text{object}}) + \mathbf{p}_o, \quad (5)$$

where $\mathbf{p}_o$ is a learned object-token embedding.

Object information is fused only into the waist, arm, and hand tokens. Let $\mathcal{I}_U$ denote these five regions. A one-layer fusion Transformer computes

$$[\tilde{\mathbf{Z}}^{\mathcal{I}_U}, \tilde{\mathbf{h}}_o] = F_{\text{fuse}}([\mathbf{Z}^{\mathcal{I}_U}, \mathbf{h}_o]; M_o), \quad (6)$$

where each selected robot token attends only to itself and the object token, and the object token attends only to itself. The base and leg tokens bypass object fusion. Importantly, the original object-independent waist token is retained for the leg-action heads, making leg actions structurally independent of direct object observations.

3.3.3 GROUPED ACTION DECODING

Seven group-specific MLP heads decode the structured features into a 41-dimensional action mean. Each head receives anatomically relevant context: leg heads use the base, corresponding leg, and object-independent waist tokens; the waist and arm heads use object-conditioned upper-body features; and each hand head uses the corresponding arm and hand features. Collectively,

$$\boldsymbol{\mu}_t = \mathcal{D}(\mathbf{Z}_t, \tilde{\mathbf{Z}}_t^{\mathcal{I}_U}). \quad (7)$$

The actor parameterizes a diagonal Gaussian policy,

$$\pi_{\theta}(\mathbf{a}_t | \mathbf{o}_t) = \mathcal{N}(\boldsymbol{\mu}_t, \text{diag}(\boldsymbol{\sigma}^2)), \quad (8)$$

with independently learned standard deviations. The 41 outputs represent normalized joint-position commands rather than joint torques. They are converted to joint-position targets using joint-specific scales and offsets and subsequently executed by joint-level PD controllers; the 12 independent hand commands are expanded through fixed mimic-joint relations.

3.3.4 SINGLE-STAGE ACTOR-CRITIC LEARNING

All actor components are jointly optimized with PPO (Schulman et al., 2017). Tracking rewards supervise the retargeted body and hand motion, while manipulation rewards supervise object interaction. The actor uses deployable observations, including noisy and delayed object measurements, whereas the critic receives privileged simulation state. DexWeave therefore learns whole-body tracking and dexterous interaction in a single training stage, without tracking-policy pretraining, teacher-student distillation, or subsequent residual-policy refinement. Further implementation and training details are provided in Appendix B.

4 EXPERIMENTS

We evaluate DexWeave at two levels on a Unitree G1 with Inspire hands: retargeting quality and closed-loop control. Section 4.1 evaluates whole-body feasibility and hand-object geometry, while Section 4.2 examines whether the retargeted references enable effective policy learning. Ablation results are reported in Appendix E.

4.1 RETARGETING EVALUATION

We evaluate DexWeave across progressively richer interaction settings: body motion in LAFAN1 (Harvey et al., 2020), whole-body object interaction without finger references in OMOMO (Li et al., 2023), and articulated hand-object interaction in GRAB (Taheri et al., 2020) and HUMOTO (Lu et al., 2025). This progression evaluates whether whole-body support remains feasible as increasingly detailed interaction geometry is introduced. Appendix A provides the details on retargeting settings, and Appendix C defines the evaluation metrics and aggregation protocol.

Whole-body feasibility and coarse contact preservation. LAFAN1 and OMOMO evaluate whole-body feasibility before introducing fine-grained finger geometry, with OMOMO additionally providing hand-object contact information. Because OMOMO does not contain observed fingertip targets, hand refinement primarily encourages contact between the robot hand and object surfaces. We compare DexWeave with OmniRetarget (Yang et al., 2026), GMR (Araujo et al., 2026), and SOMA (NVIDIA, 2026). Table 1 shows the lowest penetration and near-zero foot skating for DexWeave on both datasets. On OMOMO, contact duration reaches 0.999 and contact distance falls to 2.944 cm, compared with 0.879 and 7.677 cm for the strongest contact baseline, GMR. The contact gain is accompanied by a penetration duration of only 0.002, showing that closer hand-object alignment is achieved while retaining whole-body feasibility under the measured criteria.

Fine-grained interaction retargeting. We compare DexWeave with OmniRetarget paired with DexPilot (Handa et al., 2020) or SBR (Malate et al., 2026), each followed by arm-wrist IK that adjusts hand placement while keeping finger articulation fixed. Table 2 shows that DexWeave lowers primary-fingertip error to 4.342 mm on GRAB and 7.198 mm on HUMOTO, compared with 14.925

Table 1: **Whole-body feasibility and coarse interaction preservation on LAFAN1 and OMOMO.** Penetration and foot skating measure feasibility; contact preservation is evaluated only on OMOMO. Durations are fractions, and ~ 0 denotes a near-zero value. **Yellow** and underlined mark the best and second-best values within each dataset.

Dataset	Method	Penetration		Foot Skating		Contact Preservation	
		Duration ↓	Max Depth (cm) ↓	Duration ↓	Max Vel. (m/s) ↓	Duration ↑	Distance (cm) ↓
<i>LAFAN1</i>	OmniRetarget	<u>0.125</u>	<u>2.934</u>	0.091	<u>0.416</u>	N/A	N/A
	GMR	0.183	4.314	<u>0.010</u>	0.548	N/A	N/A
	SOMA	0.968	6.379	<u>0.115</u>	0.565	N/A	N/A
	DexWeave (Ours)	<u>~ 0</u>	<u>2.575</u>	<u>0</u>	<u>0</u>	N/A	N/A
<i>OMOMO</i>	OmniRetarget	<u>0.131</u>	<u>2.735</u>	~ 0	1.455	0.644	10.289
	GMR	0.916	3.117	0.008	0.802	<u>0.879</u>	<u>7.677</u>
	SOMA	0.746	4.163	0.034	<u>0.467</u>	0.577	16.205
	DexWeave (Ours)	<u>0.002</u>	<u>2.117</u>	<u>~ 0</u>	<u>0.355</u>	<u>0.999</u>	<u>2.944</u>

Table 2: **Fine-grained interaction retargeting on GRAB and HUMOTO.** Penetration duration is the fraction of colliding frames. Primary and secondary errors measure thumb/index and remaining-finger positions; palm error measures normal alignment. Lower is better. **Yellow** and underlined mark the best and second-best values within each dataset. Full comparison can be found in Appendix D.

Dataset	Method	Penetration		Hand Alignment		
		Duration ↓	Max Depth (cm) ↓	Pri. Err. (mm) ↓	Sec. Err. (mm) ↓	Palm Err. (°) ↓
<i>GRAB</i>	OmniRetarget + DexPilot + IK	0.212	<u>2.269</u>	16.585	<u>13.368</u>	11.555
	OmniRetarget + SBR + IK	<u>0.190</u>	2.362	<u>14.925</u>	14.659	<u>9.530</u>
	DexWeave (Ours)	<u>0.001</u>	<u>1.263</u>	<u>4.342</u>	<u>14.476</u>	<u>3.652</u>
<i>HUMOTO</i>	OmniRetarget + DexPilot + IK	0.522	3.597	32.389	31.018	14.444
	OmniRetarget + SBR + IK	<u>0.511</u>	<u>3.582</u>	<u>29.999</u>	<u>28.436</u>	<u>12.476</u>
	DexWeave (Ours)	<u>0.007</u>	<u>2.854</u>	<u>7.198</u>	<u>11.702</u>	<u>4.439</u>

and 29.999 mm for the best IK variants, while achieving the lowest penetration duration and palm error. Its secondary-fingertip error ranks first on HUMOTO and second on GRAB. These joint gains are consistent with refining the arm, wrist, and fingers as one interaction chain. Figure 4 shows representative whole-body motions and hand-object close-ups.

4.2 POLICY EVALUATION

The policy study asks whether the retargeted references can be executed under dynamics, first in body-only tracking and then in dexterous loco-manipulation. We train one policy per reference and compare actors with matched task-specific observations, actions, rewards, reset and termination rules, domain randomization, and training budgets. Reported values summarize the common final training window for each comparison; Appendices B and C provide the details on training and aggregation settings.

Motion tracking. To isolate body coordination without object conditioning, we compare DexWeave with Any2Track (Zhang et al., 2026b), GMT (Chen et al., 2025), and BeyondMimic (Liao et al., 2025) on LAFAN1 motions (Harvey et al., 2020). As shown in Table 3, DexWeave achieves a 100% completion ratio, matching BeyondMimic while outperforming all baselines in tracking accuracy. Compared with BeyondMimic, DexWeave reduces body-position error from 2.82 to 2.62 cm, torso-anchor position error from 5.19 to 4.82 cm, and anchor rotation error from 2.48° to 2.34°. Any2Track and GMT achieve lower completion ratios of 72.82% and 69.86%, respectively.

Dexterous loco-manipulation. The main control evaluation combines whole-body motion with articulated hand-object interaction on reference motions. We compare DexWeave with InterMimic

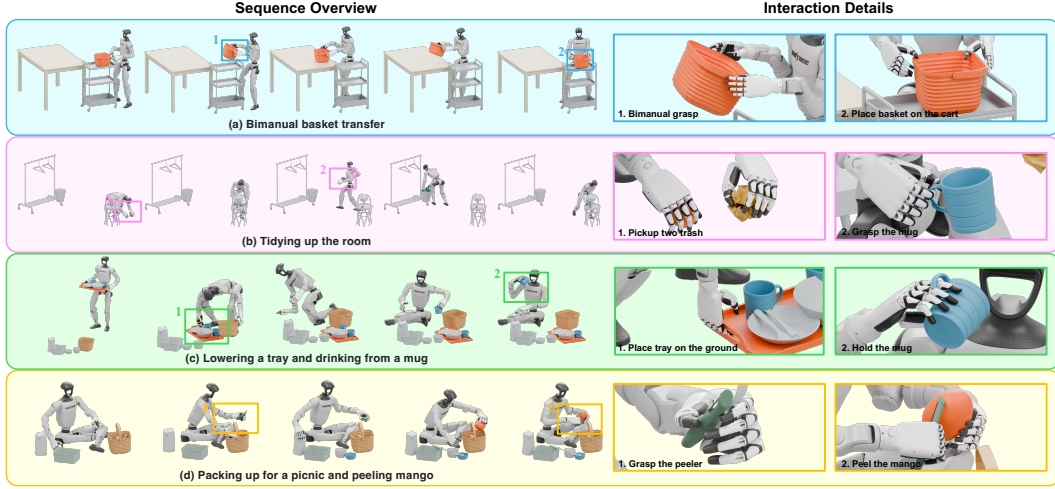


Figure 4: **Qualitative retargeting results across interaction levels.** Each row shows a sequence overview (left) and hand-object close-ups (right).

Body-only motion tracking				
Actor	Complete Ratio (%) $\uparrow$	Body pos. (cm) $\downarrow$	Anchor pos. (cm) $\downarrow$	Anchor rot. (deg) $\downarrow$
Any2Track	72.82	6.09	44.88	32.07
GMT	69.86	5.85	36.06	21.92
BeyondMimic	100	2.82	5.19	2.48
DexWeave (Ours)	100	2.62	4.82	2.34

Dexterous loco-manipulation				
Actor	Success rate (%) $\uparrow$	Body pos. (cm) $\downarrow$	Object pos. (cm) $\downarrow$	Object rot. (deg) $\downarrow$
InterMimic	57.74	6.51	3.58	7.71
Object MLP	85.00	5.69	3.84	6.16
DexWeave (Ours)	97.50	4.93	3.43	6.83

Table 3: **Policy performance under MuJoCo sim-to-sim evaluation.** Results are averaged equally over five tracking motions and five loco-manipulation references. Any2Track and GMT serve as general motion tracking baselines. Following InterMimic’s official human-model setup, we evaluate it using an SMPL-X skeleton as a cross-embodiment reference. DexWeave and Object MLP share the G1-Inspire embodiment. Completion and success rates indicate the percentage of evaluation episodes reaching the reference end or time limit. Best values are in **bold**.

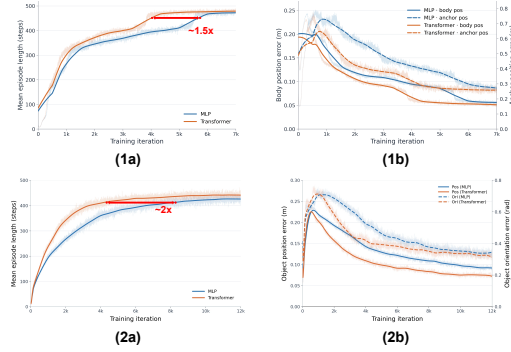


Figure 5: **Policy learning curves.** The top row shows tracking episode length (1a) and body-position (solid) and anchor-position (dashed) errors (1b); the bottom row shows loco-manipulation episode length (2a) and object-position (solid) and orientation (dashed) errors (2b). Blue denotes the MLP and orange denotes DexWeave.

(Xu et al., 2025) and an object-conditioned MLP baseline, i.e. we replace DexWeave’s Transformer network with an MLP of comparable parameter count. As shown in Table 3, DexWeave achieves the highest success rate of 97.5%, compared with 85.0% for the MLP baseline and 57.74% for InterMimic. It also attains the lowest body-position error (4.93 cm) and object-position error (3.43 cm). The MLP baseline achieves a slightly lower object-rotation error (6.16° versus 6.83°). Overall, DexWeave improves task success while maintaining more accurate body and object tracking. Figure 5 shows the corresponding training curves, further demonstrating that DexWeave converges faster than its MLP counterpart.

Sim-to-sim and hardware evaluation. We evaluate DexWeave through both sim-to-sim and sim-to-real transfer on several dexterous loco-manipulation tasks. Figure 6 shows corresponding executions across simulation environments and on a physical Unitree G1 equipped with Inspire dexterous hands. The results show that the learned policies preserve coordinated whole-body motion and dex-

terous object interaction across simulators and transfer to the physical platform, providing qualitative evidence that the human-derived references can be realized as executable robot skills.

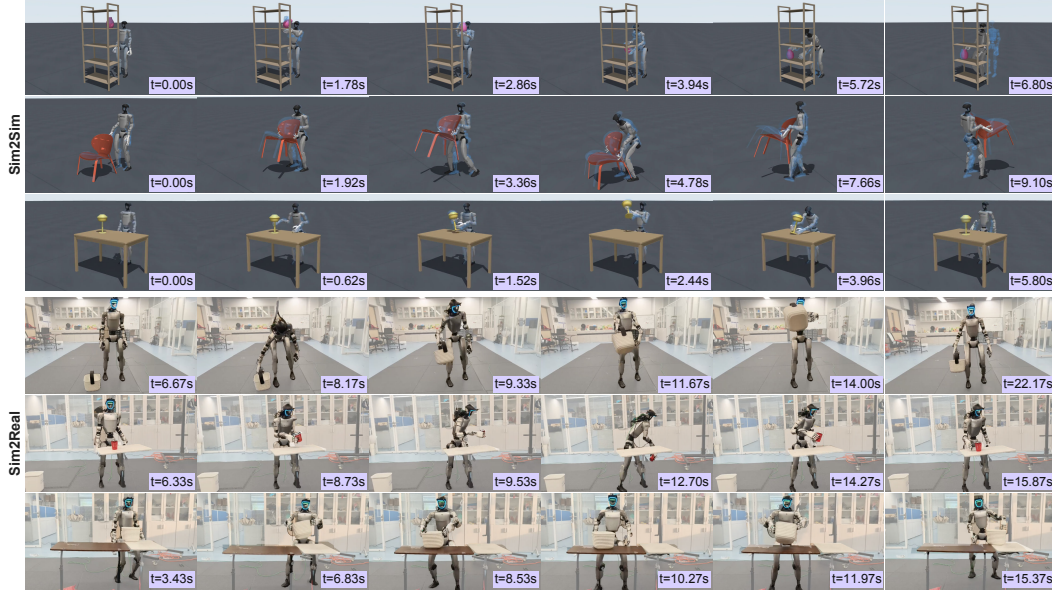


Figure 6: **Sim-to-sim and sim-to-real transfer of dexterous loco-manipulation policies.** Policies are trained in IsaacLab, evaluated in MuJoCo for sim-to-sim transfer, and deployed on a physical Unitree G1 with Inspire dexterous hands for sim-to-real transfer.

5 CONCLUSION

We introduced **DexWeave**, which transfers human demonstrations into executable dexterous humanoid loco-manipulation skills by combining interaction-consistent retargeting with anatomy-aware policy learning. Experiments in simulation and on a physical humanoid demonstrate its effectiveness in generating embodiment-specific training data for future embodied foundation models.

REFERENCES

- Arthur Allshire, Hongsuk Choi, Junyi Zhang, David McAllister, Anthony Zhang, Chung Min Kim, Trevor Darrell, Pieter Abbeel, Jitendra Malik, and Angjoo Kanazawa. Visual imitation enables contextual humanoid control. In *Conference on Robot Learning*, 2025.
- Joao Pedro Araujo, Yanjie Ze, Pei Xu, Jiajun Wu, and C. Karen Liu. Retargeting matters: General motion retargeting for humanoid motion tracking. In *IEEE International Conference on Robotics and Automation*, 2026.
- Daniel Arnström, Alberto Bemporad, and Daniel Axehill. A dual active-set solver for embedded quadratic programming using recursive LDL^T updates. *IEEE Transactions on Automatic Control*, 67(8):4362–4369, 2022. doi: 10.1109/TAC.2022.3176430.
- Zixuan Chen, Mazeyu Ji, Xuxin Cheng, Xuanbin Peng, Xue Bin Peng, and Xiaolong Wang. GMT: General motion tracking for humanoid whole-body control. *arXiv preprint arXiv:2506.14770*, 2025. doi: 10.48550/arXiv.2506.14770. URL <https://arxiv.org/abs/2506.14770>.
- Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.026. URL <https://roboticsproceedings.org/rss19/p026.html>.

-
- Ankur Handa, Karl Van Wyk, Wei Yang, Jacky Liang, Yu-Wei Chao, Qian Wan, Stan Birchfield, Nathan Ratliff, and Dieter Fox. DexPilot: Vision-based teleoperation of dexterous robotic hand-arm system. In *IEEE International Conference on Robotics and Automation*, pp. 9164–9170, 2020. doi: 10.1109/ICRA40945.2020.9197124.
- Félix G. Harvey, Mike Yurick, Derek Nowrouzezahrai, and Christopher Pal. Robust motion in-betweening. *ACM Transactions on Graphics*, 39(4), 2020. doi: 10.1145/3386569.3392480. URL <https://github.com/ubisoft/ubisoft-laforge-animation-dataset>.
- Tairan He, Zhengyi Luo, Xialin He, Wenli Xiao, Chong Zhang, Weinan Zhang, Kris M. Kitani, Changliu Liu, and Guanya Shi. OmniH2O: Universal and dexterous human-to-humanoid whole-body teleoperation and learning. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pp. 1516–1540. PMLR, 2025. URL <https://proceedings.mlr.press/v270/he25b.html>.
- Liang Heng, Yihe Tang, Jiajun Xu, Henghui Bao, Di Huang, and Yue Wang. HumDex: Humanoid dexterous manipulation made easy. *arXiv preprint arXiv:2603.12260*, 2026.
- Yingdong Hu, Haodong Zhu, Boyuan Zheng, Yihang Hu, Tong Zhang, Zunhao Chen, Junming Zhao, Ruiqian Nai, and Yang Gao. OpenHLM: An empirical recipe for whole-body humanoid loco-manipulation. *arXiv preprint arXiv:2606.22174*, 2026.
- Haoran Jiang, Jin Chen, Qingwen Bu, Li Chen, Modi Shi, Yanjie Zhang, Delong Li, Chuanzhe Suo, Chuang Wang, Zhihui Peng, and Hongyang Li. WholeBodyVLA: Towards unified latent VLA for whole-body loco-manipulation control. In *International Conference on Learning Representations*, 2026. URL <https://iclr.cc/virtual/2026/poster/10009805>.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P. Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model. In *Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pp. 2679–2713, 2025. URL <https://proceedings.mlr.press/v270/kim25c.html>.
- Jiaman Li, Jiajun Wu, and C. Karen Liu. Object motion guided human motion synthesis. *ACM Transactions on Graphics*, 42(6):1–11, 2023. doi: 10.1145/3618333. URL <https://lijiaman.github.io/projects/omomo/>.
- Sikai Li, Shuning Li, Zhenyu Wei, Yunchao Yao, Chenran Li, and Mingyu Ding. CoordDex: Coordinating body and hand priors for continuous dexterous humanoid loco-manipulation. *arXiv preprint arXiv:2606.23680*, 2026a. URL <https://arxiv.org/abs/2606.23680>.
- Zhe Li, Zhenzhe Zhang, Yangyang Wei, Wenjie Zhang, Xichen Yuan, Peiyuan Zhi, Gen Li, Xinying Guo, Fengjie Gao, Jianfei Yang, and Shanghang Zhang. ω -0: A latent predictive world action model for concurrent humanoid loco-manipulation. *arXiv preprint arXiv:2608.06375*, 2026b.
- Zhuo Li, Yiming Yao, Jim Tan, Mengjie Jing, Zhipeng Dong, and Fei Chen. WholeBodyWAM: Generalizing pre-trained world-action priors to humanoid loco-manipulation via WBC-grounded coordination. *arXiv preprint arXiv:2609.16644*, 2026c.
- Qiayuan Liao, Takara E. Truong, Xiaoyu Huang, Yuman Gao, Guy Tevet, Koushil Sreenath, and C. Karen Liu. BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion. *arXiv preprint arXiv:2508.08241*, 2025.
- Jiaxin Lu, Chun-Hao Paul Huang, Uttaran Bhattacharya, Qixing Huang, and Yi Zhou. HUMOTO: A 4D dataset of mocap human object interactions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 10886–10897, 2025. doi: 10.1109/ICCV51701.2025.01013. URL <https://jiaxin-lu.github.io/humoto/>.
- Robert Jomar Malate, Erik Bauer, Norica Bacuieti, Stefanos Charalambous, Elvis Nava, Robert K. Katzschmann, and Benedek Forrai. Smooth operator: A real-time sampling-based algorithm for kinematic hand retargeting. *arXiv preprint arXiv:2607.07491*, 2026. URL <https://arxiv.org/abs/2607.07491>.

-
- NVIDIA. SOMA retargeter. Software, 2026. URL <https://github.com/NVIDIA/soma-retargeter>.
- Chaoyi Pan, Changhao Wang, Haozhi Qi, Zixi Liu, Homanga Bharadhwaj, Akash Sharma, Tingfan Wu, Guanya Shi, Jitendra Malik, and Francois Hogan. SPIDER: Scalable physics-informed dexterous retargeting. *arXiv preprint arXiv:2511.09484*, 2025. URL <https://arxiv.org/abs/2511.09484>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. doi: 10.48550/arXiv.1707.06347. URL <https://arxiv.org/abs/1707.06347>.
- Wandong Sun, Luying Feng, Baoshi Cao, Yang Liu, Yaochu Jin, and Zongwu Xie. ULC: A unified and fine-grained controller for humanoid loco-manipulation. *arXiv preprint arXiv:2507.06905*, 2025. URL <https://arxiv.org/abs/2507.06905>.
- Omid Taheri, Nima Ghorbani, Michael J. Black, and Dimitrios Tzionas. GRAB: A dataset of whole-body human grasping of objects. In *European Conference on Computer Vision*, pp. 581–600, 2020. doi: 10.1007/978-3-030-58548-8_34. URL <https://grab.is.tue.mpg.de/>.
- Jielin Wu, Shenzhe Yao, Guanqi He, Xiaohan Liu, Zhaoqing Zeng, Xiangrui Jiang, Han Yang, Wentao Zhang, and Hang Zhao. TopoRetarget: Interaction-preserving retargeting for dexterous manipulation. *arXiv preprint arXiv:2606.16272*, 2026.
- Chendong Xin, Mingrui Yu, Yongpeng Jiang, Zhefeng Zhang, and Xiang Li. Analyzing key objectives in human-to-robot retargeting for dexterous manipulation. *IEEE Robotics and Automation Practice*, 1:29–34, 2026. doi: 10.1109/RAP.2026.3656110.
- Sirui Xu, Hung Yu Ling, Yu-Xiong Wang, and Liang-Yan Gui. InterMimic: Towards universal whole-body control for physics-based human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12266–12277, 2025. URL <https://sirui-xu.github.io/InterMimic/>.
- Lujie Yang, Xiaoyu Huang, Zhen Wu, Angjoo Kanazawa, Pieter Abbeel, Carmelo Sferrazza, C. Karen Liu, Rocky Duan, and Guanya Shi. OmniRetarget: Interaction-preserving data generation for humanoid whole-body loco-manipulation and scene interaction. In *IEEE International Conference on Robotics and Automation*, 2026.
- Kevin Zakka. Mink: Python inverse kinematics based on MuJoCo, 2026. URL <https://github.com/kevinzakka/mink>. Software, version 1.1.0.
- Yuanhang Zhang, Yifu Yuan, Prajwal Gurunath, Ishita Gupta, Shayegan Omidshafiei, Ali-akbar Agha-mohammadi, Marcell Vazquez-Chanlatte, Liam Pedersen, Tairan He, and Guanya Shi. FALCON: Learning force-adaptive humanoid loco-manipulation. In *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pp. 265–281, 2026a. URL <https://proceedings.mlr.press/v331/zhang26a.html>.
- Zhikai Zhang, Jun Guo, Chao Chen, Jilong Wang, Chenghuai Lin, Yunrui Lian, Han Xue, Zhenrong Wang, Maoqi Liu, Jiangran Lyu, Huaping Liu, He Wang, and Li Yi. Track any motions under any disturbances. In *IEEE International Conference on Robotics and Automation*, 2026b. URL <https://arxiv.org/abs/2509.13833>.
- Jia Zheng, Teli Ma, Yudong Fan, Zifan Wang, Shuo Yang, and Junwei Liang. MotionWAM: Towards foundation world action models for real-time humanoid loco-manipulation. *arXiv preprint arXiv:2606.09215*, 2026.
- Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog,

Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183, 2023. URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.

A RETARGETING DETAILS

A.1 REFERENCE PREPROCESSING AND DATASET TARGETS

Time and coordinate conventions. All human, robot, and object trajectories are represented at 50 Hz ($\Delta t = 0.02$ s), preserving their original timing. Using zero-based frame indices, output frame k corresponds to the fractional source index $k f_{\text{src}}/50$, where f_{src} is the source frame rate. Rotations are interpolated on $\text{SO}(3)$. A shared heading alignment and ground-height translation are applied to all trajectories and world-space targets. The original scene scale is preserved by default ($s_{\text{scene}} = 1$). When real-world interaction targets exceed the robot’s reach, we adjust the scene scale only as needed, applying $\mathbf{p}' = \mathbf{a} + s_{\text{scene}}(\mathbf{p} - \mathbf{a})$ to the human and all objects about a common ground anchor $\mathbf{a}$. Object mesh dimensions use the same scale factor; their orientations are unaffected by scaling. The resulting object trajectories remain fixed throughout retargeting.

Interaction-dependent targets. For sequences with articulated hand observations, let $\mathbf{p}_{t,s}^{\text{morph}}$ denote the morphology-adapted wrist target and $\mathbf{p}_{t,s}^{\text{scene}}$ the corresponding wrist position in the preprocessed scene, for hand $s \in \{L, R\}$. The wrist position target is

$$\hat{\mathbf{p}}_{t,s} = (1 - \alpha_{t,s})\mathbf{p}_{t,s}^{\text{morph}} + \alpha_{t,s}\mathbf{p}_{t,s}^{\text{scene}}. \quad (9)$$

The morphology-adapted target follows the torso. The scene target follows the preprocessed demonstration in the shared world frame and is not modified by retargeting. The blend applies only to wrist positions, while wrist orientation targets retain their mapped values. The interaction weight $\alpha_{t,s} \in [0, 1]$ is determined from persistent hand–object proximity, using a 5 cm threshold and a nominal persistence/transition interval of 4/30 s, discretized to seven frames at 50 Hz. GRAB and HUMOTO additionally use source finger-contact labels to activate scene tracking, with a smooth proximity-based ramp during approach. The same interaction weight balances world-frame finger-tip alignment and wrist-local hand-shape preservation during refinement.

Dataset-specific targets. GRAB and HUMOTO provide articulated hand observations for fingertip and hand-orientation fitting. Their evaluation masks are defined directly from source finger-contact labels, independently of the smoothed target activations used during optimization. OMOMO provides body–object motion without observed fingertip targets, while its SMPL-H joint centers are therefore not treated as fingertip observations. Instead, refinement retains the observed wrist positions and initialized hand orientations, and uses robot-surface contact targets in place of fingertip tracking. LAFAN1 provides body-only motion, which fingers remain in a neutral configuration, and refinement updates only the arm and wrist joints. The corresponding refinement objectives and contact-construction details are provided in Appendix A.3.

A.2 DECOUPLED BODY–HAND INITIALIZATION SETTINGS

A.2.1 BODY OPTIMIZATION AND SUPPORT CONSTRAINTS

The body solver in Eq. 1 optimizes the floating base and non-finger joints, with finger joints fixed. Table 4 summarizes its soft objectives. The feasible set is

$$\mathcal{Q}_t^B = \left\{ \mathbf{q}^B : \begin{array}{l} \mathbf{q}^- \leq \mathbf{q}_{\text{act}}^B \leq \mathbf{q}^+, \\ b_s(\mathbf{q}^B) \geq 0 \quad (s \in \{L, R\}), \\ \mathbf{c}_{s,xy}(\mathbf{q}^B) = \mathbf{a}_{t,s} \quad (s \in \mathcal{S}_t), \\ d_c(\mathbf{q}^B) \geq -\epsilon_c \quad (c \in \mathcal{C}_t^B) \end{array} \right\}, \quad (10)$$

where $\mathbf{q}_{\text{act}}^B$ contains the actuated body coordinates. The elbow constraint preserves the physical bending side through

$$b_s = \frac{\mathbf{n}_s^\top [(\mathbf{p}_w - \mathbf{p}_e) \times (\mathbf{p}_e - \mathbf{p}_{\text{sh}})]}{\|\mathbf{p}_w - \mathbf{p}_e\|_2 \|\mathbf{p}_e - \mathbf{p}_{\text{sh}}\|_2},$$

with wrist, elbow, and shoulder positions $\mathbf{p}_w$, $\mathbf{p}_e$, and $\mathbf{p}_{\text{sh}}$, and the robot elbow’s world-frame bending axis $\mathbf{n}_s$.

Fixed-foot detection in the source. The sole-stick mask is inferred once from the unscaled source toe-base positions, ankle positions, and foot headings on the 50 Hz time grid, before morphology adaptation. Toe height above the inferred support surface, ankle lift relative to its stable-contact height, horizontal and vertical toe speeds, and foot yaw rate determine continuous contact, support, stance, settling, motion, swing, liftoff, and landing scores in $[0, 1]$, with causal smoothing. For ground support, the stationarity gate requires horizontal toe speed at most 0.15 m/s and absolute yaw rate at most $30^\circ/\text{s}$. Contact and transition scores additionally distinguish a planted foot from a slowly moving airborne foot.

The fixed-foot flag requires this stationarity gate, contact score at least 0.50 or support score at least 0.55, aggregate support strength at least 0.55, and motion, swing, liftoff, and landing scores all below 0.25. Aggregate support strength is the maximum of the support, stance, and settling scores and the release-attenuated contact score $c(1 - r^3)$, where c is the contact score and r is the maximum motion, swing, and liftoff score. Only contiguous runs lasting at least 0.12 s (six frames at 50 Hz) are retained. Each retained run receives a separate episode identifier, and the active feet define $\mathcal{S}_t$. The mask and identifiers remain fixed during optimization.

Support height is inferred per episode to include feet planted on raised surfaces. Candidate raised plateaus lie more than 12 cm above the global floor, persist for at least 0.12 s, and have a 5th–95th percentile height range at most 2.5 cm; their acceptance also checks descent onto the surface and subsequent departure. An accepted raised-support episode supplies the stationarity gate for the interaction datasets. LAFAN1 additionally requires horizontal and vertical toe speeds at most 0.30 and 0.25 m/s and yaw rate at most $100^\circ/\text{s}$. The same contact, transition, and persistence conditions still apply.

Hard horizontal support constraints. At each frame, the body solver reads the fixed source mask to acquire, retain, or release a foot constraint. The constrained robot toe point $\mathbf{c}_s(\mathbf{q})$ is located at $[0.14, 0, -0.03]^\top \text{ m}$ in the G1 foot frame. At episode entry t_e , the solver stores its position $\mathbf{c}_{e,s}$ from the entry robot configuration and the corresponding mapped source contact pose $(\hat{R}_{t_e,s}, \hat{\mathbf{p}}_{t_e,s})$. The first output frame establishes the initial anchors. During an episode, the desired horizontal anchor is

$$\mathbf{a}_{t,s} = \Pi_{xy} \left[\hat{\mathbf{p}}_{t,s} + \hat{R}_{t,s} \hat{R}_{t_e,s}^\top (\mathbf{c}_{e,s} - \hat{\mathbf{p}}_{t_e,s}) \right], \quad (11)$$

where Π_{xy} selects the horizontal coordinates. LAFAN1 uses its independently constructed source toe target with an episode-constant horizontal offset acquired from the entry configuration. Thus the constraint preserves the robot’s contact offset relative to source support motion; a stationary source target gives a constant world-space anchor. The anchor is released when the source sole-stick flag becomes inactive.

For every $s \in \mathcal{S}_t$, the equality $\mathbf{c}_{s,xy}(\mathbf{q}) = \mathbf{a}_{t,s}$ enters both body IK passes as a hard QP constraint. At iterate $\mathbf{q}^{(k)}$, its linearization is

$$J_{s,xy}(\mathbf{q}^{(k)}) \Delta \boldsymbol{\xi} = \mathbf{a}_{t,s} - \mathbf{c}_{s,xy}(\mathbf{q}^{(k)}), \quad (12)$$

where $\Delta \boldsymbol{\xi}$ is the tangent-space configuration increment. Joint and collision bounds are enforced in the same solve. Toe forward kinematics is checked again after integration; nonlinear step correction controls the equality residual with a nominal per-coordinate tolerance of 10^{-4} m . Finite outputs that retain a larger residual are recorded as support-quality violations and remain subject to output-based skating evaluation. The equality constrains only horizontal toe position: height follows the support surface and nonpenetration constraints, while foot orientation is tracked through soft objectives. Subsequent hand fitting and upper-body refinement leave the floating base and legs fixed, preserving the attained support configuration.

Collision boundaries. The signed separation d_c is positive outside collision. The set $\mathcal{C}_t^B$ includes ground and support-surface boundaries, filtered body self-collision pairs, and enabled body–object boundaries; detailed finger geometry is excluded. We use $\epsilon_c = 0$, except for the GRAB and HUMOTO non-arm body–object check, which allows $\epsilon_c = 0.02 \text{ m}$. Collision and support-equality tolerances are 10^{-5} m and 10^{-4} m , respectively. These optimization settings are distinct from evaluation thresholds.

Table 4: **Body-optimization objectives.** Position and orientation-axis terms use squared Euclidean and squared angular errors, respectively; posture terms act on joint coordinates. Weights are per landmark, axis, or joint, with $k = 10^3$. Foot and knee weights are listed before support/contact modulation. Lengths are in meters and angles in radians unless stated otherwise.

Term	Target	Weight or distance scale
<i>Body and support tracking</i>		
Pelvis pose	World position; mapped $x/y/z$ axes.	82k; (2.4, 2.4, 3.2)k
Torso pose	World position; mapped axes.	11k; (1.05, 1.05, 1.45)k
Shoulder / elbow positions	World-frame shoulders; torso-relative elbows.	4.2k ; 1.2k
Wrist pose	Interaction-dependent position; mapped axes.	240k; (2.1, 1.7, 0.9)k
Foot positions	Support-aware sole center; heel and toe positions; sole-corner heights.	115k; 48k each; 260k each
Foot orientation	Forward axis; sole normal.	1.8k; 2.8k
Knee position	World positions during ground-contact motion.	115k
<i>Temporal regularization ($t > 1$)</i>		
Root orientation	Previous root correction transported by the source rotation change.	(1.2, 1.2, 1.6)k
Torso / elbow corrections	Previous pose/position corrections transported with the current reference.	$w_{\text{track}}\tau_B/\Delta t$
Non-arm joint posture	Previous leg-twist, other leg, waist, and remaining non-arm joint angles.	1.20, 0.16, 0.12, 0.045
Arm joint posture	Previous arm/wrist angles.	$\tau_B h_j/(2\Delta t)$
<i>Posture and bending priors</i>		
Leg bending	Source bending direction/branch; optional knee-flexion targets outside ground episodes.	$420o^2$; 420
Arm joint margins	Neutral posture in joint-margin coordinates: wrist, shoulder/elbow, and other arm joints.	1.20, 0.03, 0.025
Initial body posture	Neutral non-arm joint angles at the first frame.	0.018
<i>Soft collision clearance</i>		
Self / ground clearance	Body geometry and arm proxies; self-clearance margin of 5 mm and zero penetration into the ground or active support plane.	35 mm
Object clearance	Surface/material distance; zero margin at contact patches and 3 mm elsewhere.	3 mm for fine body geometry; 35 mm for coarse/arm geometry
Exact object penetration	Robot collision convexes against object meshes; smooth zero-penetration penalty.	0.1 mm

Axis-weight triples follow x, y, z . Here $\tau_B = 0.1$ s, giving $\tau_B/\Delta t = 5$ at 50 Hz; w_{track} is the corresponding tracking weight. The quantity h_j is the Gauss–Newton curvature of the arm-margin term at the previous configuration, and $o \in [0, 1]$ measures source leg-bend observability. Clearance residuals are based on $[m_c - d_c]_+$ and the listed distance scales, where $[x]_+ = \max(x, 0)$. Exact penetration uses a smoothed boundary transition. Terms are enabled only when the required references or geometry are available.

A.2.2 HAND CODEBOOK AND CONTINUOUS FITTING

Each hand is parameterized by six independent drivers. Dependent joints follow the mimic relation $q_j = a_j\theta_{d(j)} + b_j$. Driver bounds are intersected with the bounds implied by all dependent joints. Forward kinematics yields wrist-local descriptors of finger directions, accumulated bending angles, reach-to-chain-length ratios, and normalized inter-finger distances. Retrieval uses the mean squared distance over observed components selected by the validity mask $\mathbf{m}_t$:

$$D_{tn} = \frac{\|\mathbf{m}_t \odot (\mathbf{g}_t^{s,H} - \mathbf{g}_n^{s,R})\|_2^2}{\sum_j m_{t,j}}, \quad (13)$$

$$\hat{\theta}_t^s = \frac{\sum_{n \in \mathcal{I}_t^s} \exp(-D_{tn}/\tau) \theta_n^s}{\sum_{n \in \mathcal{I}_t^s} \exp(-D_{tn}/\tau)}.$$

Here $\mathcal{I}_t^s$ contains the $K = 12$ nearest entries and $\tau = 0.06$.

Starting from $\hat{\theta}_t^s$, continuous fitting combines wrist-local fingertip and chain position residuals, a posture prior informed by the retrieved configurations and observed finger shape, and first- and

Table 5: **Hand-initialization settings.** Fitting weights are per landmark or driver and multiply squared position or driver-coordinate errors. Position targets are wrist-local; lengths are in meters and angles in radians. Here $k = 10^3$.

Term	Target and activation	Weight or setting
<i>Codebook and retrieval</i>		
Driver sampling	Six mimic-feasible drivers: thumb yaw/pitch and four finger flexions; interval sampling followed by forward kinematics.	$\{0.15, 0.50, 0.85\}$; $N_B = 3^6 = 729$ per hand
Finger descriptors	Wrist-to-tip, chain-root, and up to three segment unit vectors; accumulated bending and tip reach divided by chain length.	Up to $15 + 2$ components per finger
Inter-finger distances	Selected tip-to-tip distances normalized by the mean wrist-to-tip radius.	6 pairs; at most 91 descriptor components overall
Masked retrieval	Observed-component distance and exponential driver blending in Eq. equation 13.	$K = 12$; $\tau = 0.06$
<i>Geometric fitting</i>		
Fingertip positions	Observed thumb, index, and remaining fingertips.	$4.5k f_{\text{tip}} r p$
Finger-chain positions	Observed j_1, j_2, j_3 , and fingertip landmarks.	$2.25k f_{\text{chain}} \ell_j p$
<i>Posture and support priors</i>		
Shape posture	Six-driver reference informed by observed geometry and retrieved configurations.	$170(1 + 3\sigma + 6\eta)$
Support hold	Shape-reference driver angles during stable hand support.	$374 \alpha c_{\text{sup}}$
Target hold	Previous driver angles while holding an interaction target.	$510 \eta c_{\text{tgt}}$
Thumb support	Thumb yaw/pitch when a support pitch target is available.	$1.2k \beta(0.78, 1.35)$
<i>Temporal regularization</i>		
First-order smoothness	Previous driver configuration, after the first frame.	$460(1 + 4\sigma + 10\eta)$
Second-order smoothness	Extrapolated driver configuration $2\theta_{t-1} - \theta_{t-2}$.	$180(1 + 3\sigma + 7\eta)$

Thumb/index/other-finger factors are $(1.45, 1.25, 0.78)$ for f_{tip} and $(1.25, 1.08, 0.82)$ for f_{chain} . For j_1, j_2, j_3 , and tip, respectively, ℓ_j takes 0.40, 0.72, 0.95, and 0.85; non-thumb j_3 landmarks receive an additional factor of 0.35. The pass factor is $p = 0.72$ for coarse fitting and $p = 1.18$ for fine fitting; $r = 1.45$ for the optional fingertip-focused solve and $r = 1$ otherwise. The strengths σ , η , α , and β denote support stability, target hold, support hold, and thumb support, respectively. The per-driver factors $c_{\text{sup}} = (2, 2, 2.6, 2.8, 2.8, 2.8)$ and $c_{\text{tgt}} = (1.8, 2.1, 2.8, 3, 3, 3)$ follow the six-driver order. Coarse and fine IK use eight iterations each with damping 0.5, subject to driver bounds and mimic relations. Unavailable observations and inactive support terms contribute no residual; fingers remain neutral without articulated hand observations.

second-order driver smoothness. Optimization updates only the independent drivers within their bounds, with dependent joints reconstructed through the mimic relations. Table 5 details the codebook construction, retrieval, and continuous-fitting settings. The fitted hands and body trajectory together form the decoupled initialization $\bar{\mathbf{q}}_{1:T}$. Sequences without articulated hand references use neutral fingers at this stage.

A.3 COUPLED UPPER-BODY INTERACTION REFINEMENT

Refinement jointly optimizes 14 arm/wrist joints and 12 independent hand drivers for OMOMO, GRAB, and HUMOTO. LAFAN1 updates only the arm/wrist joints and keeps the fingers neutral. The floating base, legs, and waist retain their initialized trajectories, and the preprocessed object trajectories remain unchanged.

Interaction objectives. For articulated-hand inputs, let $\mathbf{p}_{t,sf}$ and $\mathbf{u}_{t,sf}$ denote the world-frame and wrist-local positions of fingertip f on hand s . Their corresponding targets are $\mathbf{y}_{t,sf}$ and $\mathbf{v}_{t,sf}$, and $\ell_{t,s}$ denotes the target hand scale. Using the interaction weight $\alpha_{t,s}$ from reference construction, fingertip alignment is defined as

$$\mathcal{L}_{\text{tip}} = \sum_{s,f} \frac{\omega_f}{\ell_{t,s}^2} \left[\alpha_{t,s}^2 \|\mathbf{p}_{t,sf} - \mathbf{y}_{t,sf}\|_2^2 + (1 - \alpha_{t,s})^2 \|\mathbf{u}_{t,sf} - \mathbf{v}_{t,sf}\|_2^2 \right]. \quad (14)$$

Table 6: **Coupled-refinement objectives.** Position weights are divided by $\ell_{t,s}^2$ and joint-coordinate weights by squared feasible ranges. Activation factors are applied as described in the text. Angular residuals use radians.

Term	Target and activation	Weight or setting
<i>Interaction alignment</i>		
Thumb / index fingertips	World-frame and wrist-local targets in Eq. equation 14.	2.0 each
Other fingertips	Corresponding targets for the middle, ring, and little fingers.	0.20 each
Hand orientation	Palm normal, wrist-forward direction, and lateral axis.	0.55, 0.42, 0.12
<i>Position and posture anchors</i>		
Wrist / palm / elbow positions	Soft position anchors; wrist and palm anchors relax with increasing interaction activation.	0.08 / 0.06 / 0.0125
Arm posture	Posture prior in joint-limit-aware coordinates.	0.60
Hand posture	Initialized hand-driver configuration.	0.035
<i>Temporal regularization ($t > 1$)</i>		
Arm correction	Previous correction relative to the current initialization.	$\lambda_U = 80.0$
Hand drivers	Previous independent driver configuration.	$\lambda_H = 0.40$
<i>Soft object clearance</i>		
Hand-object clearance	Clearance residual with an additional penalty inside material.	1.80
Noncontact margin	Desired separation for noncontact samples.	3 mm

World-frame alignment preserves interaction placement, whereas wrist-local alignment preserves free-hand shape. Orientation terms use squared angles between corresponding palm-normal, wrist-forward, and lateral directions. Soft position and posture anchors in $\mathcal{L}_{\text{other}}$ are listed in Table 6. Wrist and palm position weights are additionally multiplied by $(1 - \alpha_{t,s})^2$, relaxing their anchors as scene alignment becomes dominant.

Temporal regularization. Let $\delta_t^U = \mathbf{q}_t^U - \bar{\mathbf{q}}_t^U$ denote the arm correction relative to the initialization. We regularize changes in this correction and in the hand drivers:

$$\mathcal{L}_{\text{temp}} = \lambda_U \|D_U(\delta_t^U - \delta_{t-1}^U)\|_2^2 + \lambda_H \sum_s \|D_H(\theta_t^s - \theta_{t-1}^s)\|_2^2, \quad (15)$$

where D_U and D_H normalize each coordinate by its feasible range. The arm term preserves the previous correction relative to the current initialization rather than anchoring the arm to its previous pose. The hand term directly smooths driver motion. Both terms are omitted at the first frame.

Feasibility and collision handling. The refinement feasible set is

$$\mathcal{Z}_t = \left\{ \mathbf{z} : \begin{cases} \mathbf{q}^- \leq \mathbf{q}_{\text{act}}(\mathbf{z}; \bar{\mathbf{q}}_t) \leq \mathbf{q}^+, \\ q_j = a_j \theta_{d(j)} + b_j, \\ b_s \geq 0 \quad (s \in \{L, R\}), \\ d_c \geq 0 \quad (c \in \mathcal{C}_t^U) \end{cases} \right\}, \quad (16)$$

where $\mathbf{q}_{\text{act}}(\mathbf{z}; \bar{\mathbf{q}}_t)$ contains the reconstructed actuated coordinates, with all inactive coordinates fixed to their initialized values. Mimic relations and elbow-bending constraints use the definitions from body-hand initialization. The set $\mathcal{C}_t^U$ contains active upper-limb self-collision boundaries and enabled ground constraints. HUMOTO and OMOMO enable ground checks for the full arm, palm, and finger geometry; LAFAN1 excludes palm and finger geometry for the reason of non-finger dataset. Adjacent and allowed robot pairs are filtered out, and fine self-collision uses URDF collision convexes. Ground and self-collision checks use a numerical tolerance of 10^{-5} m.

Object clearance and contact recovery are soft objectives, rather than constraints in $\mathcal{C}_t^U$. For a sampled object distance d_i and clearance margin m_i , the clearance residual is $r_i = [m_i - d_i]_+ + [-d_i]_+$, where $[x]_+ = \max(x, 0)$. The second term increases the penalty inside object material. Material penetration and visible-surface proximity use distinct geometric queries.

Table 7: **Actor and critic observations for dexterous loco-manipulation.** Dimensions count scalar inputs before encoding. The last column specifies regional assignments for actor inputs and feature composition for critic inputs.

Observation	Dimension	Input details
<i>Actor observations</i>		
Reference body and hand joint positions	41	Corresponding joint groups
Reference non-finger joint velocities	29	Legs, waist, arms
Reference-anchor position history relative to torso	7×3	Base
Reference-anchor orientation relative to torso	6	Base
Base angular velocity	3	Base
Current independent joint positions	41	Corresponding joint groups
Current non-finger joint velocities	29	Legs, waist, arms
Previous applied normalized action	41	Corresponding joint groups
Robot subtotal	211	Eight regional tokens
Current and reference object poses and their coordinate difference	$9 + 9 + 9$	Object token
Actor total	238	
<i>Privileged critic observations (training only)</i>		
Reference joint states	82	41 positions and 41 velocities
Anchor pose	9	Position and orientation
Tracked-link poses	216	24 link positions and orientations
Base velocities	6	Linear and angular velocities
Current joint states	82	41 positions and 41 velocities
Previous actions	41	Previous action vector
Object features	27	Current and reference poses and their coordinate difference
Critic total	463	Independent value-network input

Actor regional input dimensions are 30 for the base, 30 per leg, 15 for the waist, 35 per arm, and 18 per hand. The actor omits finger velocities, whereas the critic includes velocities for all 41 independent joint coordinates.

A.4 NUMERICAL IMPLEMENTATION

Frames are processed in temporal order using damped differential IK with Mink (Zakka, 2026) and DAQP (Arnström et al., 2022). The nominal coarse/fine iteration limits are 22/30 for body optimization and 8/8 for hand fitting; refinement uses 40 iterations per solve. Body damping is 0.30/0.25 for the coarse/fine passes, and hand fitting and refinement use 0.50. The body stage additionally uses constrained SLSQP for arm optimization. Feasibility recovery may require additional iterations. After each integration step, nonlinear geometry is re-evaluated, inactive coordinates remain fixed, and mimic relations are enforced in both configuration and tangent coordinates.

B POLICY IMPLEMENTATION AND TRAINING

Policies are trained in IsaacLab Simulator for a Unitree G1 equipped with Inspire RH56DFQ hands, with a separate policy for each retargeted reference. The network architecture is shown in Figure 3. The loco-manipulation comparison evaluates DexWeave and an object-conditioned MLP on different tasks. Within each reference, the actors share observations, action interfaces, rewards, privileged critics, randomization, and episode settings.

B.1 OBSERVATIONS AND NETWORK CONFIGURATION

Actor observations. The loco-manipulation actor receives 238 inputs: 211 robot features and 27 object features, as detailed in Table 7. The torso serves as the reference anchor. Current joint positions are expressed relative to the default configuration, whereas reference joint positions remain absolute targets. Base angular velocity is expressed in the base frame, while anchor and object features use the current torso frame. The anchor position history contains seven samples spanning 120 ms at 50 Hz. Finger velocities are omitted from the actor input.

Object observations. For a pose $T = (R, \mathbf{p})$, define $r_6(R) = [R_{11}, R_{12}, R_{21}, R_{22}, R_{31}, R_{32}]^\top$ and $\phi(T) = [\mathbf{p}^\top, r_6(R)^\top]^\top$. The object observation is

$$\mathbf{x}_o = \begin{bmatrix} \phi(T_o) \\ \phi(T_o^{\text{ref}}) \\ \phi(T_o) - \phi(T_o^{\text{ref}}) \end{bmatrix} \in \mathbb{R}^{27}. \quad (17)$$

Both poses are expressed in the same measured torso frame. The final nine components are coordinate-wise differences of the pose representations, rather than a relative rigid-body transform. The reference object trajectory remains prescribed in the scene and the simulated object moves dynamically.

Privileged critic observations. During training, the critic receives privileged observations, including finger velocities, tracked-link poses, and base linear velocity that are unavailable to the actor. Robot and object states are obtained directly from simulation without the observation noise or delay applied to the actor. The critic inputs are concatenated and processed by an independent value network. Table 7 lists the complete critic observations.

Network configuration. Each regional encoder has one 128-unit hidden layer and produces a 128-dimensional token. The object encoder receives the separate 27-dimensional object observation. The robot Transformer contains two blocks and the object-fusion Transformer contains one. Each block uses four attention heads, a feed-forward width of 256, residual connections, layer normalization, ELU activations, and no dropout. Each action head has one 128-unit hidden layer. The independent value network has hidden widths (512, 256, 128). Empirical observation normalization is applied to network inputs. The object-conditioned MLP baseline has three hidden layers of width 621 and uses the same critic.

Auxiliary motion tracking task. The body-only actor uses 157 inputs, six body tokens, and 29 actions. It retains one anchor-position sample and omits hand observations, hand actions, and object conditioning. Its privileged critic receives 286 inputs. The attention-mask ablations change only the permitted connections described in Sec. E.

B.2 ACTION INTERFACE AND REWARD DESIGN

Joint-position control. The policy outputs 41 normalized commands: 12 for the legs, three for the waist, 14 for the arms and wrists, and 12 for the independent hand drivers. These commands define joint position targets as

$$\mathbf{q}_t^{\text{cmd}} = \mathbf{q}^{\text{default}} + \mathbf{s} \odot \mathbf{a}_t, \quad (18)$$

where $s_j = 0.25\tau_j^{\text{max}}/k_{p,j}$ for body joints and $s_j = 0.25$ for hand drivers. Joint-level PD controllers execute the targets, with nominal hand-load gravity compensation applied to the waist and arms. Finger gains are $k_p = 20 \text{ N m/rad}$ and $k_d = 1 \text{ N m s/rad}$, with torque and velocity limits of 2 N m and 5 rad/s. Leg and waist target velocities are limited to 16 rad/s, except for hip yaw at 20 rad/s. The previous-action observation records the normalized target after this command processing.

The 12 independent hand commands are expanded through fixed mimic relations into 24 articulated finger joints, yielding 53 simulated joints in total. The thumb’s intermediate and distal joints use ratios of 1.6 and 2.4 relative to the pitch driver; the other four fingers use unit-ratio coupling. Reference observations and action decoding use consistent anatomical joint groups.

Reward terms. Tracking rewards take the form $r_k = \exp(-E_k/\sigma_k^2)$, where E_k is a squared position, joint-angle, velocity, or rotation-angle error. Orientation errors use geodesic angles on $\text{SO}(3)$. For multi-link and joint terms, squared errors are averaged over the selected elements before applying the exponential. Body-pose rewards use the aligned reference defined in Appendix C. Hand poses are evaluated relative to their respective wrists, and wrist–object terms compare relative poses. Table 8 gives the tracking weights, error scales, and control penalties.

The undesired-contact penalty excludes permitted foot and hand contacts. Second-order action penalties are computed from policy outputs before command processing. For body-only tracking, we omit hand and object rewards and the second-order leg-action penalty. The torso position

Table 8: **Reward configuration for dexterous loco-manipulation.** Tracking rewards use the listed weights and error scales σ , and penalty terms use direct coefficients. Lengths are in meters and angles in radians.

Term	Weight	Scale σ
<i>Whole-body tracking</i>		
Global torso position / orientation	1.5/1.5	0.20/0.20
Aligned body position / orientation	1.0/1.0	0.30/0.40
Global body linear / angular velocity	1.0/1.0	1.0/3.14
Body joint positions	1.0	0.40
<i>Hand and object interaction</i>		
Independent hand-driver positions	0.3	0.50
Wrist-relative hand position / orientation	0.3/0.3	0.10/0.40
Wrist-object relative position / orientation	1.2/1.0	0.20/0.40
Object position / orientation	1.2/1.0	0.15/0.90
<i>Control regularization</i>		
Action first difference, squared norm	-0.10	-
Policy-output second difference, waist / legs	-0.05/ - 0.01	-
Joint-limit violation	-10.0	-
Undesired contacts above 1 N	-0.10	-

Table 9: **PPO and simulation settings.**

Setting	Value	Setting	Value
Parallel environments	4,096	Physics / control frequency	200 / 50 Hz
Steps per environment	24	Samples per update	98,304
Learning epochs / minibatches	5 / 4	Initial learning rate	10^{-3}
PPO clipping parameter	0.2	Target KL divergence	0.01
Discount γ / GAE λ	0.99 / 0.95	Value-loss coefficient	1.0
Maximum gradient norm	1.0	Initial action standard deviation	1.0
Entropy coefficient, tracking / manipulation	0.005 / 0.002		

and orientation rewards use weights of 1.0 and 0.75, with corresponding error scales of 0.20 m and 0.30 rad.

B.3 POLICY OPTIMIZATION AND DOMAIN RANDOMIZATION

PPO training. All actor components and the privileged critic are optimized jointly with PPO using the settings in Table 9. Each update partitions the collected transitions into four minibatches of 24,576 samples. The learning rate is adapted using the target KL divergence, and the value loss uses clipping. Loco-manipulation metrics are aggregated over the common training window specified in Appendix C.

Dynamics and observations domain randomization. During training, we randomize contact and inertial properties, actuation and control parameters, and actuator delays. External disturbances are applied as velocity perturbations at randomized intervals. Actor observations additionally include measurement noise and delays, with object poses sampled at a fixed frequency. The critic receives the corresponding states from the same randomized simulation without the observation noise or delays applied to the actor. Table 10 summarizes the randomization models and ranges, together with the fixed measurement settings.

B.4 EPISODE INITIALIZATION AND TERMINATION

Reference-phase initialization. The default initialization scheme uses equal proportions of first-frame and sampled-phase starts. Robot and object poses are initialized at the same reference phase, and their velocities follow the reference without added impulses. For sampled starts, phases are drawn using 70% uniform sampling and 30% failure-adaptive sampling. The adaptive weights use

Table 10: **Domain randomization and measurement settings.** Randomized quantities are sampled uniformly over the listed ranges; $\pm a$ denotes $[-a, a]$. Multiplicative factors are relative to nominal values. Fixed settings are marked explicitly.

Quantity	Model	Range or value
<i>Contact and inertial properties</i>		
Static friction	Coefficient	[0.4, 1.8]
Dynamic friction	Coefficient	[0.35, 1.5]
Restitution	Fixed coefficient	0
Foot contact offset	Distance parameter	[5, 20] mm
Body mass	Multiplicative	[0.9, 1.1]
Hand payload	Multiplicative	[0.8, 1.25]
Torso center of mass	Position offset	$\pm(0.025, 0.05, 0.05)$ m
Hand center of mass	Position offset	± 0.03 m per axis
<i>Actuation and control</i>		
Motor strength	Multiplicative	[0.8, 1.2]
Joint armature	Multiplicative	[0.8, 1.2]
PD stiffness	Multiplicative	[0.85, 1.15]
PD damping	Multiplicative	[0.75, 1.25]
Default joint position	Additive offset	± 0.02 rad
Joint friction	Parameter value	[0, 0.08]
Actuator delay	Physics-step delay	0–2 steps (0–10 ms)
<i>External disturbances</i>		
Push interval	Event interval	[1.5, 3.0] s
Horizontal velocity	Velocity disturbance	± 0.4 m/s in x and y
Roll/pitch angular velocity	Velocity disturbance	± 0.15 rad/s
Yaw angular velocity	Velocity disturbance	± 0.25 rad/s
<i>Actor observation noise</i>		
Joint position	Additive noise	± 0.01 rad
Joint velocity	Additive noise	± 0.5 rad/s
Base angular velocity	Additive noise	± 0.2 rad/s
Anchor position	Additive noise	± 0.05 m
Anchor orientation	Representation noise	± 0.05 per component
Object position	Translation perturbation	± 0.01 m
Object orientation	Rotation perturbation	± 0.035 rad
<i>Actor measurement timing</i>		
Anchor-position delay	Measurement delay	[0, 120] ms
Object measurement rate	Fixed sampling rate	30 Hz
Object measurement latency	Measurement delay	[20, 60] ms

Static and dynamic friction use consistent coefficients. Torso center-of-mass offsets follow the x, y, z axes. Anchor orientation noise acts on rotation-representation components, whereas object orientation perturbations are specified in radians. The critic receives simulation observations without the noise or measurement delays applied to the actor.

the square root of phase-wise failure rates, with phase-bin probabilities capped at 0.28. Sampled-start episodes use 0.5–1.5 s of pre-roll and last at most 10 s.

Termination conditions. Episodes end at the reference end or the applicable time limit, or earlier when a failure condition is met. Both loco-manipulation and body-only tracking terminate when the torso or end-effector height error exceeds 0.25 m. For loco-manipulation, additional failure thresholds are 0.50 m for torso horizontal position error and 0.70 rad for torso orientation error. Object position and orientation errors are limited to 0.30 m and 2.0 rad, respectively. Body-only tracking additionally terminates when the vertical projected-gravity discrepancy exceeds 0.8 and has a time limit of 500 control steps.

C EVALUATION DEFINITIONS

C.1 RETARGETING METRICS

Evaluation setup. Each method is evaluated using its corresponding robot and hand geometry and physical sampling interval. Source contact and support schedules are temporally aligned with the output frames. For the modular baselines, DexPilot or SBR initializes finger articulation on the OmniRetarget body trajectory. The additional IK variants optimize the two seven-joint arm–wrist chains using world-frame wrist and fingertip position objectives, with the fingers, floating base, waist, and legs fixed. The refinement ablation compares DexWeave’s initialized and refined outputs on matched sequences, verifying that all inactive body coordinates and object trajectories remain identical.

Penetration. Penetration is recomputed from the saved output configuration at every frame. For each applicable component $c \in \{\text{ground}, \text{self}, \text{object}\}$, let $p_{t,c} \geq 0$ be the deepest detected penetration at frame t . Ground and filtered self-penetration use each method’s MuJoCo G1 collision geometry, excluding structurally allowed pairs, while palm and finger geometry are excluded for LAFAN1. OMOMO evaluates robot collision-surface points against its material signed distance field, whereas GRAB and HUMOTO compare robot URDF collision convexes against the original object triangle meshes. The component thresholds are $\tau_{\text{ground}} = \tau_{\text{self}} = 1$ cm and $\tau_{\text{object}} = 2$ cm for all interaction datasets. LAFAN1 has no object component. For a sequence of T output frames,

$$D_{\text{pen}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\exists c : p_{t,c} > \tau_c]. \quad (19)$$

Here $\mathbf{1}[\cdot]$ is the indicator function. A frame is counted once even if several components violate their thresholds. Raw penetration depths are retained without subtracting the thresholds. These evaluation thresholds are applied to the returned trajectory independently of the optimizer’s clearance objectives, feasibility tolerances, or convergence status.

Foot skating. The fixed-foot mask and episode identifiers are derived from source motion as detailed in Appendix A.2.1, then sampled at each method’s output times. They are independent of the optimized robot toe speed. At $t > 1$, let $\mathcal{V}_t$ contain the feet whose source sole-stick flag is active at both $t - 1$ and t with the same episode identifier. The first frame and transitions into a new episode provide no valid within-episode displacement for that foot. Using the robot toe point defined in Appendix A.2.1, its horizontal speed is

$$v_{t,s} = \frac{\|\mathbf{c}_{s,xy}(\mathbf{q}_t) - \mathbf{c}_{s,xy}(\mathbf{q}_{t-1})\|_2}{\Delta t}, \quad (20)$$

where Δt is the evaluated output’s physical sampling interval. A frame is classified as skating when any $s \in \mathcal{V}_t$ has $v_{t,s} > 0.30$ m/s. The sequence-level duration is

$$D_{\text{skate}} = \frac{\sum_{t=2}^T \mathbf{1}[\exists s \in \mathcal{V}_t : v_{t,s} > 0.30 \text{ m/s}]}{\sum_{t=2}^T \mathbf{1}[\mathcal{V}_t \neq \emptyset]}. \quad (21)$$

The denominator counts frames with at least one eligible support foot, counting double support once. Sequences with a zero denominator have undefined skating metrics and are excluded from skating averages. The 0.30 m/s threshold classifies output violations; source support activation and hard optimization constraints use the separate rules in Appendix A.2.1.

Contact preservation and hand alignment. Evaluation frames are determined by source interactions, independently of robot contact. OMOMO detects contact using unsigned distance threshold to the visible object surface. Source detection uses the minimum distance over five reconstructed distal-finger proxies per hand and robot detection uses samples of its native hand collision surface. GRAB and HUMOTO instead select each hand’s interaction frames using positive source finger-contact labels.

On OMOMO, contact preservation is the fraction of source-contact frames in which either robot hand contacts any active object part, without requiring the same hand or part as in the source. Contact distance averages unsigned distances from all five robot fingertips to the nearest active object surface on frames when the corresponding source hand is in contact. SOMA uses five fixed surface proxies on its rigid rubber hand. For finer alignment on GRAB and HUMOTO, primary and secondary errors measure world-frame fingertip distances for the thumb, index and remaining fingers, respectively. Palm error measures the angle between semantic palm normals. Targets come from the source scene before morphology blending or optimization adjustments.

C.2 POLICY METRICS AND AGGREGATION

Geometric errors. Let a denote the torso anchor and b a tracked link. Body-position error uses the yaw-aligned, height-preserving reference transform of BeyondMimic (Liao et al., 2025):

$$R_{\Delta,t} = R_z(\text{yaw}(R_{t,a}(R_{t,a}^{\text{ref}})^\top)),$$

$$\tilde{\mathbf{p}}_{t,b}^{\text{ref}} = \begin{bmatrix} p_{t,a,x} \\ p_{t,a,y} \\ p_{t,a,z}^{\text{ref}} \end{bmatrix} + R_{\Delta,t} (\mathbf{p}_{t,b}^{\text{ref}} - \mathbf{p}_{t,a}^{\text{ref}}). \quad (22)$$

This transform aligns the reference yaw and horizontal anchor position with the robot while preserving the reference height. The body-position error at frame t is

$$e_{\text{body},t} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \|\mathbf{p}_{t,b} - \tilde{\mathbf{p}}_{t,b}^{\text{ref}}\|_2. \quad (23)$$

For body-only tracking, $\mathcal{B}$ contains 14 links: the pelvis, torso, and bilateral hip-roll, knee, ankle-roll, shoulder-roll, elbow, and wrist-yaw links. Loco-manipulation additionally includes ten hand landmarks.

Anchor and object errors are evaluated directly in the world frame, without reference alignment. Position error is the Euclidean distance between corresponding positions, and orientation error is

$$d_{\text{SO}(3)}(R_1, R_2) = \arccos\left(\text{clip}\left(\frac{\text{tr}(R_1^\top R_2) - 1}{2}, -1, 1\right)\right). \quad (24)$$

Episode statistics. Episode length is measured in control steps, with a maximum of 500 steps for body-only tracking. An episode is considered complete if it reaches the reference end or the prescribed time limit without early failure termination. Completion ratio for body-only tracking and success rate for loco-manipulation report the percentage of completed evaluation episodes. For sampled-phase starts, completion refers to the evaluated segment rather than necessarily the entire reference.

Evaluation protocol and aggregation. Body-only tracking and loco-manipulation policies are trained for 8k and 12k PPO iterations, respectively. During the final 1k iterations, policy checkpoints are saved every 100 iterations. For each reference sequence, every saved checkpoint is evaluated in MuJoCo five times using different evaluation seeds, with policy weights fixed throughout evaluation. All tabulated policy results are computed from these evaluation runs rather than training logs. For each metric, we first average across the five evaluation runs for each checkpoint, then across checkpoints within each sequence, and finally equally across reference sequences within each task setting.

D ADDITIONAL RETARGETING RESULTS

Table 11 gives the full comparison, including baselines with and without arm-wrist IK. Adding IK substantially lowers fingertip error but increases penetration duration on both datasets. DexWeave improves primary-fingertip alignment and reduces penetration duration relative to both variants.

Table 11: **Complete fine-grained interaction retargeting results on GRAB and HUMOTO.** Penetration duration is the fraction of colliding frames. Primary and secondary errors measure thumb/index and remaining-finger positions; palm error measures normal alignment. Lower is better. **Yellow** and underlined mark the best and second-best values within each dataset.

Dataset	Method	Penetration		Hand Alignment		
		Duration ↓	Max Depth (cm) ↓	Pri. Err. (mm) ↓	Sec. Err. (mm) ↓	Palm Err. (°) ↓
GRAB	OmniRetarget + DexPilot	<u>0.183</u>	2.123	209.353	207.745	52.122
	OmniRetarget + DexPilot + IK	0.212	2.269	16.585	13.368	11.555
	OmniRetarget + SBR	0.184	<u>2.120</u>	210.524	212.985	52.122
	OmniRetarget + SBR + IK	0.190	2.362	<u>14.925</u>	14.659	<u>9.530</u>
	DexWeave (Ours)	0.001	1.263	4.342	<u>14.476</u>	3.652
HUMOTO	OmniRetarget + DexPilot	<u>0.370</u>	3.638	292.886	278.574	47.815
	OmniRetarget + DexPilot + IK	0.522	3.597	32.389	31.018	14.444
	OmniRetarget + SBR	<u>0.370</u>	3.638	292.750	283.381	47.815
	OmniRetarget + SBR + IK	0.511	<u>3.582</u>	<u>29.999</u>	<u>28.436</u>	<u>12.476</u>
	DexWeave (Ours)	0.007	2.854	7.198	11.702	4.439

Table 12: **Retargeting-stage ablation.** Decoupled is the body-and-hand initialization; refined is the final output.

Dataset	Stage	Penetration		Hand Alignment		
		Duration ↓	Max Depth (cm) ↓	Pri. Err. (mm) ↓	Sec. Err. (mm) ↓	Palm Err. (deg) ↓
GRAB	Decoupled	0.157	1.778	36.033	36.995	0.189
	Refined	0.001	1.263	4.342	14.476	3.652
HUMOTO	Decoupled	0.204	2.568	32.703	26.090	0.159
	Refined	0.007	2.854	7.198	11.702	4.439

E ABLATION STUDY

Effect of upper-limb refinement. We compare the decoupled initialization and refined output on matched GRAB and HUMOTO sequences to isolate the contribution of coupled refinement (Table 12). Refinement reduces primary-fingertip error by 87.9% on GRAB and 78.0% on HUMOTO, together with lower secondary-fingertip errors. Penetration duration decreases from 0.157 to 0.001 and from 0.204 to 0.007, respectively. These improvements are accompanied by larger palm-normal errors on both datasets. On HUMOTO, Max Depth also increases from 2.568 to 2.854 cm. Because this metric averages frame-wise maximum depths only over violating frames, the increase reflects the severity of the remaining violations rather than their frequency. Overall, refinement improves fingertip alignment and reduces penetration frequency, with a trade-off in palm orientation and conditional penetration depth on HUMOTO.

Effect of the attention masks. We relax each attention mask separately, keeping the remaining architecture and training settings fixed. Dense M_R enables unrestricted attention among all eight robot-region tokens. Bidirectional M_O enables dense attention among the object token and five upper-body tokens, retaining the lower-body bypass. Table 13 shows that the full design achieves the highest completion ratio of 100% and the lowest body-position error of 4.58 cm. Dense M_R

Table 13: **Policy-mask ablation.** Results are averaged over two reference motions using checkpoints saved every 100 iterations during the final 1,000 training iterations, with five evaluation seeds per checkpoint and reference. Dense M_R and bidirectional M_O each relax one attention mask.

Attention setting	Complete Ratio (%) ↑	Body pos. (cm) ↓	Object pos. (cm) ↓	Object rot. (deg) ↓
M_R dense	91.30	4.79	3.62	8.23
M_O bidirectional	99.71	4.63	3.88	9.19
M_R and M_O (DexWeave)	100	4.58	3.81	8.47

yields lower object-position and orientation errors (3.62 cm and 8.23° , compared with 3.81 cm and 8.47° for DexWeave), but its completion ratio falls to 91.30%. Bidirectional M_O achieves a similar completion ratio of 99.71%, with higher body-position and object-pose errors than the full design. These results support directed attention for episode completion and body tracking, without showing a uniform advantage in object-pose accuracy.